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Planning for cost-effective underground space use in urban areas

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ABSTRACT: The issues surrounding planning for a cost effective use of underground space in urban areas are discussed. Having an understanding of the most appropriate uses for underground space in a particular city, how these might change as the city grows and how the geology, topography and existing land use affect the viability of underground uses is important. The problems of initiating such planning efforts in a timely fashion also are discussed.

1. INTRODUCTION

A colloquium was held in July 1995 in Minnesota, USA to review some of the basic tenets regarding the future use of the underground (Sterling 1995). The participants were only drawn from the U.S. but the professional backgrounds of the participants included planners; architects; public works, transportation and geotechnical engineers; and a former city mayor. The participants also represented a variety of institutional and topical interests including the U.N., the World Bank, the National Research Council's various programs on infrastructure and the constructed environment, and several federal and city government entities. A reasonable proportion of the participants had only a limited involvement in considering or working on underground projects in the past.

Despite the mix of disciplines and backgrounds, a clear consensus emerged that the use of underground space and the construction of underground facilities represents a part of our constructed environment that will grow **proportionally more important** with time. Land use and environmental issues were identified as the principal driving forces behind this trend and large cities, particularly in areas of rapid urbanization, were identified as the places where the needs and opportunities are the greatest. In rural areas and in areas of severe urban sprawl, the underground will be less used because of cost and reduced need but even in these areas there are trends to place more utility services and more service facilities underground to improve the surface environment. Should urban sprawl become insupportable from a regional transportation perspective, as has happened in Los Angeles, then some form of local redensification of the urban area is envisaged to allow mass transportation possibilities to develop. The underground could form an important component of such redeveloped areas.

The above consensus was felt to be the first step in examining the future use of the underground and what can be done to make future uses more effective and more efficient. If use of the underground was not expected to increase in relative terms then it would have been more difficult to make compelling arguments about the need for national or city-wide action to redress current problems. The consensus then became the basis for discussions about the implications of an increasing role for the underground in terms of public policy, current planning practices, and research needs. There are many aspects of underground technology that offer substantial potential for improvement that could reduce construction costs and physical and financial risks. These reduced costs would encourage more underground construction, increase the market size and promote further private research. Coupled to technological

innovation, however, must be the administrative procedures which permit underground projects to be developed in a reasonable way and which prevent a destruction of the future potential of the underground resource in an area by poorly planned initial uses.

In discussing the planning for cost effective use of the underground, it is worth repeating the summary findings of the colloquium:

1. Use of the underground will grow relatively more important in the future (increasing as a proportion of our infrastructure expenditures).
2. In order to plan cities effectively for the future, the underground must be understood, considered as an alternative for many functions, and its uses properly planned.
3. Underground planning and space utilization can make an increasing contribution to:
   - Enhanced livability of cities
   - Improved public health
   - Multiple use and reuse of land through synergistic surface and subsurface development
   - Rehabilitating/re-energizing older core cities
   - Efficient transport and provision of services
4. There is a growing rationale for increased underground space utilization because of:
   - Urban growth worldwide
   - Need for infrastructure renewal in older cities
   - Need for basic infrastructure in developing countries
   - Congested and deteriorating utility and transportation corridors
   - Current improvements in underground construction technology
5. Steps to improve both the technologies available and the administrative handling of underground space utilization would reap large rewards for society in the future. Increased use of the underground, appropriately developed and managed, will provide increased mobility of people, goods and services, improved environmental quality, and can solve unique national needs such as isolating radioactive waste.
6. Use of the underground is becoming increasingly competitive with surface strategies for providing services and facilities, because:
   - Improved technologies provide more alternatives and improve existing alternatives
   - The public is insisting on including non-economic criteria rather than using lowest construction (or even life-cycle) financial cost in choosing project alternatives
7. An increase in competitiveness of the underground option and benefits derived from underground use can be accelerated by the concerted action of government, industry, and the research and education communities.

2. CHOOSING TO BUILD UNDERGROUND

In order to choose properly between building a facility underground or on the surface, between various design and construction options for the facility, and whether to build the facility at all, it is necessary to understand the full range of costs and benefits for the facility as are outlined in the next section. Life cycle costing and social cost accounting are important elements in this assessment. This is not an easy task -- the impacts on maintenance, longevity and operating costs of various design options may not be readily available and the value of improvements in the safety, the environment and quality of life for individuals have been assessed at widely different levels. It is also difficult to assess the performance and state of deterioration of existing underground systems or to properly address potential differences in the performance of underground systems during infrequent but high-risk natural disasters.

It is important to present an honest story in this regard. Building structures underground can be expensive and prone to unexpected conditions that raise costs and can cause surface disruption. However, such problems can be reduced and more cost effective technologies and designs developed when the problem is viewed as a major opportunity to be tackled.

The competitiveness of underground construction is increasing with time. This is good news but costs are still high and design issues as well as improved technologies must be addressed to reach the most cost effective solutions.

Under some conditions, it is reasonable to examine use of the underground as one that involves the choice between a surface facility and one placed underground. In an increasing number of cases, however, public opposition to surface construction of major new developments and infrastructure are forcing a choice between building underground or not building at all. By a similar argument, it is not necessary to try to make sweeping choices between building up or down in dense urban areas. Building dense aboveground developments requires at a minimum the underground utility infrastructure needed to serve the developments. In fact, the ability to create effective underground facilities provides the basis for creating the added value above ground. This has been termed creating an "address" and has been used effectively to spur redevelopment in projects such as the La Defense project in Paris, Crystal City near Washington D.C. and Battery Park City in New York. In the last case, the Battery Park City Authority not only placed the infrastructure but also owned the land that benefited from the increase in value as the development was realized.

3. CHARACTERISTICS OF UNDERGROUND SPACE USE

It is not easy to summarize the attributes of underground facilities across a wide range of facility uses and project locations. Advantages to an underground location in one circumstance may be a disadvantage in another. At best the potential benefits and potential drawbacks can be outlined and these must then be evaluated for their applicability in different geographic, climatic, geological and site conditions. The accompanying table provides a checklist of parameters for consideration. A fuller description and discussion of each issue can be found in the source publication (Carmody and Sterling 1993).

Three major rationales for the use of underground space can be drawn from the table. These are:

1. Land Use and Locational Advantages: The use of underground space allows a facility to be built in a location where a surface facility is not possible either because of lack of surface space or because building a surface facility in that location is not acceptable to the community. There are many types of facilities that are best or necessarily placed in locations where their physical presence on the surface is unwanted – public utilities and care parks for example. There is also often the need to separate conflicting transport activities or to provide easy connections among them. The distributions of pedestrians around major train stations and bus/train interchanges are examples of this type of need. Grade separation of uses is often desired and planning one use underground generally impacts far less on the existing community.
2. Physical and Isolational Advantages: The ground is massive and opaque and provides a variety of advantages in terms of isolation and protection – either of the underground facility from the surface or the surface from the nuisance of the underground facility. These are important aspects in designing facilities with a low environmental impact.
3. Topographic Freedom Advantages and Crossings: In hilly or mountainous areas, the use of tunnels (and bridges) improves or makes feasible various transport options such as roads, railways, canals, etc. In urban areas, several levels of transport facilities can be brought together in the important city transport hubs. Tunnels also are an important option in river and harbor crossings.

There are many drawbacks to underground facilities – some of these may preclude the use of an underground facility and some may require the use of special design features. Fiscal drawbacks are important and usually include: a high initial cost to construct the facility, a high operational cost for underground transport facilities, and a concern for the risk of unforeseen conditions that may delay construction and increase costs. Psychological reservations about going underground and dislike of cramped windowless spaces are also problems that must be ameliorated as much as possible by using frequent access to natural light and view, spacious and well laid out interior designs, and high quality finish materials.

Table 1 Benefits and Drawbacks of Underground Facilities
(Carmody and Sterling 1993)

Major Issues	Sub Category	Potential Benefits	Potential Drawbacks
Physical and Institutional Issues	Location	Proximity Lack of surface space Service provision Status	Unfavorable geology Uncertain geology
	Isolation	Climatic *thermal, severe weather, fire, earthquake* Protection *noise, vibration, explosion, fallout, industrial accident* Security *access, protected surfaces* Containment *hazardous materials, hazardous processes*	Climatic *thermal, flooding* Communication Human issues *psychological acceptability, psychological concerns, fire safety, personal safety*
	Preservation	Aesthetics *visual impact, interior design* Environmental *natural landscape, ecology, run-off* Materials	Aesthetics *visual impact, building service, skilful design* Environmental *site degredation, drainage, pollution*
	Layout	Topographic freedom 3-dimensional planning	Ground support Span limitations Access limitations Adaptability Sewage removal
	Institutional		Easement acquisition Permits Building code Investment uncertainty
Life Cycle Cost	Initial Cost	Land cost savings Construction savings *no structural support, weather independent, scale* Sale of excavated materials or minerals Savings in specialized design features	Confined work conditions Ground support Limited access Ground excavation, transportation, and disposal Cost uncertainty *geological, contractual, institutional delays*
	Operating Cost	Maintenance Insurance Energy use	Equipment/materials access Personnel access Ventilation and lighting Maintenance and repair
Societal Issues		Land use efficiency Transportation and Circulation efficiency Energy conservation Environment/aesthetics Disaster readiness National security Less construction disruption	Environmental degradation Permanent changes Embodied energy

It is important is that planning options for a project or community development should not be unnecessarily limited by lack of understanding of more unfamiliar options and/or preconceptions about their cost/difficulty. There have been radial changes in the technology underground construction in the past few decades that are having a significant impact on the cost, speed, and local impact of underground construction. It is also important to note that many of the difficulties of underground construction in urban areas are caused by prior underground uses, poorly planned (and even less well recorded).

4. PUBLIC, PROFESSIONAL AND LEGISLATIVE ATTITUDES

When a major departure from the typical way of constructing facilities in a city or region is proposed, there are many barriers that must be overcome. One of these is overcoming the inertia that pushes for common, well-understood approaches even when a new approach conceptually has value. The forms of this inertia may include:

1. Avoidance of risk by politicians, planners and consultants by choosing a "standard" approach
2. Resistance related to the desire to keep design solutions within the experience of the consultants who may be involved in the planning phases of the work or that may be influential in the local community
3. Unreasonable concerns among the public about the impacts or desirability of underground options

These issues are in addition to the real comparison of benefits and drawbacks and to the difficulty of quantifying many of the long-term benefits of underground construction.

This is not to say that politicians, planners, designers, or the public are the problem. Champions for underground solutions usually involve members of at least one or more of these groups that are seeking better solutions to new projects or urban problems. For example, the public often may force consideration of an underground option to a new road or rail project over the concerns of the project designers as to the cost of the option. In this way, an underground location may be a solution that offers a compromise for a project unwanted by a local community – the community feeling that its concerns have been heard and settling for what they feel is the best that they can get. Starting off with an underground alternative may not always smooth the path to public acceptance. Since many projects that serve the urban area as a whole are not wanted at all by the local community, protests against the project will still occur and there is less opportunity for radical concessions if the project is to move forward.

5. PLANNING FRAMEWORK

The choice to build an underground transit system, a new form of underground industrial or energy storage facility, or an underground shopping center in a city center is rarely a smooth series of decisions that lead to the preferred underground solution. Many projects are placed underground as a compromise between opponents and proponents of a surface or elevated construction project. These are often late-stage planning or design decisions that result in adapting existing plans rather than selecting sites, routes or layouts that would favor low cost underground construction.

Problems also can exist even when the potential for underground construction has been identified well in advance. The discussions often follow the sequence below:

1. The planners/politicians understand that this is a useful idea but there is not an immediate project that would benefit from being placed underground so they do not want to spend the funds now to plan the use of the underground.
2. A developer/institution/government department has identified a need or opportunity for a new facility. Once such a project becomes a real possibility, the project backers want to proceed as quickly as possible so that the need can be met and/or before funding can be withdrawn from the project.

3. Even if the project designers and backers are conceptually receptive to the idea of an underground facility, they may not be able to proceed immediately because of concerns about how the project will fit into the overall framework of underground space and land use in the area. Because of time and funding concerns, they do not want to wait until a planning study is completed and hence choose the conventional option.
4. The need for planning is restated but now there is again no urgent need to do the planning work.

Even when extensive planning work for underground space use was carried out as was the case for the University of Minnesota (UofMN) (Fairhurst et al. 1975) and the Minneapolis-St. Paul (Twin Cities) area (Sterling and Nelson, 1982), variations of the above scenario were repeated over several decades. Despite the rising and fading of interest in underground projects and several missed opportunities, there have been tangible results, however. They just took much longer to materialize than was anticipated by those involved in the initial studies. An approximate timeline for use of underground space in the Twin Cities area is:

Early 1900s	Construction of utility tunnels in the favorable St. Peter Sandstone formation Construction of mushroom growing caves and beer storage caves in the same formation
Mid 1900s	Excavation of caves for use of the sandstone for glass making at the local Ford plant
Late 1960s	First conceptual proposals for use of the favorable geological conditions under the Minneapolis campus of the University of Minnesota for convenient, climate-controlled parking structures
Early 1970s	Planning for underground mass transit in the Twin Cities area Conference on urban uses of underground space related to the Twin Cities geology and climate Planning study for underground space use on the UofMN Minneapolis campus
1977	Opening of Williamson Hall on UofMN campus – 2-story underground building emphasizing energy efficiency and preservation of open-space on campus
1982	Opening of Civil and Mineral Engineering building on the UofMN campus as a demonstration of the use of mined space in the St. Peter Sandstone for building and industrial purposes
Early 1980s	Planning studies for industrial use of mined space in Minneapolis and St. Paul Planning for light rail system in the Twin Cities area with underground tunnel in central Minneapolis
Late 1990s	Planning and construction of a mined space library regional archives facility at the University of Minnesota with the facility planned to act as a gateway to further underground space use Planning and construction of an initial light rail system for the Twin Cities area with an underground section at the airport

There are many other examples of excellent planning work for the use of underground space around the world but these also usually show over a period of decades a rather confused interaction of identification of opportunities, proactive planning for long-term effective use, and reactive planning to control expanding and poorly planned uses.

6. TRIGGERS FOR USE

The above discussions illustrate that having an identified benefit and a generally receptive legislative and planning framework does not mean that underground solutions will be adopted as new projects are proposed. Both industry and government uses will need specific "triggers" to initiate the first example projects. For example, what caused the first sewage treatment plant to be built underground in Scandinavia? Now it is considered a normal practice and there has to be a good reason to revert to a surface plant design but there was obviously a stage at which it had to be "sold" as an idea. There needed to be sufficient rationale to overcome few or no

existing examples to follow and there had to be champions of the idea in the design, construction and administrative communities.

Interest in the potential of improved land use and low environmental impact led to a 1983-84 study of the use of underground space for storage and disposal in the U.K. funded by the Department of the Environment (Dames and Moore, 1984). Table 2 shows the table developed to illustrate the relative weight of issues that were considered to be triggers relevant to underground storage and disposal.

By a similar identification and ranking of the issues that affect the use of such underground caverns by industry in other countries, one could conclude whether such uses would be principally determined by site location, geology, environmental controls, etc. and for what types of facilities would financial incentives, planning incentives or environmental restrictions prove the most effective in encouraging underground construction.

It is important to conduct both national strategic planning as well as local geology-specific planning for many types of industrial facilities so that an understanding of the best locations for various types of underground facilities is developed as well as more specific conceptual and schematic designs that depend on local geology, topography and transportation network, e.g. Godard, 1996; ITA 1987, 1990, 1995, 1997 1999; Monnikhof et al, 1998; Watanabe, 1990.

7. REFERENCE PROJECTS

Because of the difficulty of creating generic cost-benefit studies of underground facilities versus their surface counterparts, the availability of reference projects can be very important in providing case histories of similar decisions faced by other cities and regions and tangible examples of the successes and difficulties experienced when the underground solution was chosen. It is much easier for politicians, planners and designers to make a choice about a new type of facility for their region when they have been able to observe such facilities first hand and to experience the internal environment, observe the integration of the facility into the surface environment, and question the operators/occupants/users of the facility.

8. EFFECTIVE USE OF UNDERGROUND SPACE

The use of the underground in cities has a history of almost a total lack of long-range planning. Often there are no planning tools available to control the use of underground space under either public or private land. This first come, first served approach to the use of underground space and a lack of regard for the potential future value of underground space increases the costs of major underground projects later in the life of a city as well as lessening their usability. For example, near-surface utilities force pedestrian tunnels to be set deeper beneath streets creating longer transitions and more stairs to climb. Inefficient underground space use at major urban centers and intermodal facilities causes difficulties in the interface of new systems. A current study by Brown and Root for the U.K. Highways Agency is looking at conceptual designs of intermodal facilities using underground space as a means to optimize their cost effectiveness in future planning and design. Recent studies in Singapore (TUST 1999) have identified the best geological locations for various types of facilities of urban and national strategic interest and developed the conceptual designs that allow an effective discussion of the costs, benefits and impacts of the facilities.

Fixing basic geometric planning decisions early and following through on these in design can help simplify the integration of surface uses, transportation and utilities and encourage the effective use of underground space in serving those needs. For example, in the planning of a new suburban center (Joondalup) near Perth, decisions on the basic hierarchy for road, rail and retail levels for the center were made during the mid 1970s. This included the decision to bring rail access into the center below ground, crossing beneath adjacent roadways and allowing direct rail access to the heart of the center. The realization of these plans did not occur until the 1990s but road, rail and development plans for the area were able to proceed on a common basis of understanding in the intervening years.

Table 2 Triggers To Use Of Underground Storage Caverns By Industry In The U.K.
(Dames and Moore 1984)

Advantages		Hydrocarbon		Urban Industrial Storage	Cold & Frozen	Waste	Energy	Water & Sewage	Agri-culture
		Liquid	Gas						
Economic	Capital cost	□	■	□	□	□	□	□	□
	Operating cost	□	□	◘	◘	□	□	□	□
Mitigation of Environmental Impact		◘	◘	□	□	◘	■	■	■
Access to prime site		■	■	■	■	■	■	■	■
Financial support for underground mining		■	□	■	■	■	■	□	■
Financial support for management of surface amenities		■	■	■	■	□	□	■	□
Civil defence		■	■	■	■	□	□	□	■
Controlled environment for sensitive operations		□	□	■	□	□	□	◘	□

■ Major Advantage ◘ Some Advantage □ Minor Advantage

9. CONCLUSIONS

Planning for cost effective use of underground space in urban areas depends on a number of ingredients for its culmination in successful projects:

1. A framework for understanding the types of facilities and the locations that offer the best potential for effective underground space use in a city or region usually provided by a broad-based planning study for the area.
2. Resolution of legislative or administrative barriers to the effective use of underground space (if they exist) so that the initial projects are not delayed or abandoned due to the time delays and uncertainties of resolving such issues.
3. Identification or creation of local projects that will serve as physical examples of the advantages of the use of underground space and the quality of interior spaces for public uses.
Identification of reference projects in other cities that provide examples of successful solutions in similar physical and economic conditions.
4. A understanding of the long-term impacts of intrusive infrastructure development and environmental degredation to provide balance to arguments about the first cost of underground solutions versus surface or elevated solutions
5. Local and regional planning for the use of underground space that will maximize the future benefit of the space and allow speedy decisions or planning criteria to be made available for new projects.
6. Targeting the use of underground space in terms of allowing a better functioning and more livable urban area as a whole, i.e. the increased use of underground space is not a goal in itself.

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